In [24]:
```
#Sameer Kapse
#Roll_NO:TI60
#Assignment 4
#Perform the following operations using Python on the Facebook metrics data sets
#a. Create data subsets
#b. Merge Data
#c. Sort Data
#d. Transposing Data
#e. Shape and reshape Data
```

Out[24]:
```
'#Sameer Kapse \n#Assignment 4\nPerform the following operations using Python on t
he Facebook metrics data sets\na. Create data subsets\nb. Merge Data\nc. Sort Data
\nd. Transposing Data\ne. Shape and reshape Data'
```

In [5]:
```
import pandas as pd
import numpy as np
df=pd.read_csv("dataset_Facebook.csv",sep=";")
df
```

Out[5]:

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	Lifetime Post Total Impressions	Lifetime Engaged Users	C
0	139441	Photo	2	12	4	3	0.0	2752	5091	178	
1	139441	Status	2	12	3	10	0.0	10460	19057	1457	
2	139441	Photo	3	12	3	3	0.0	2413	4373	177	
3	139441	Photo	2	12	2	10	1.0	50128	87991	2211	
4	139441	Photo	2	12	2	3	0.0	7244	13594	671	
...	...	...	...	...	...	...	...	...	...	...	
495	85093	Photo	3	1	7	2	0.0	4684	7536	733	
496	81370	Photo	2	1	5	8	0.0	3480	6229	537	
497	81370	Photo	1	1	5	2	0.0	3778	7216	625	
498	81370	Photo	3	1	4	11	0.0	4156	7564	626	
499	81370	Photo	2	1	4	4	NaN	4188	7292	564	

500 rows × 19 columns

In [6]:
```
#shape
df.shape
```

Out[6]:
```
(500, 19)
```

In [10]:
```
#Create subset1
df1=df[['Page total likes','Category','Post Month','Post Weekday']].loc[0:15]
df1
```

Out[10]:

	Page total likes	Category	Post Month	Post Weekday
0	139441	2	12	4
1	139441	2	12	3
2	139441	3	12	3
3	139441	2	12	2
4	139441	2	12	2
5	139441	2	12	1
6	139441	3	12	1
7	139441	3	12	7
8	139441	2	12	7
9	139441	3	12	6
10	139441	2	12	5
11	139441	2	12	5
12	139441	2	12	5
13	139441	2	12	5
14	138414	2	12	4
15	138414	2	12	3

In [11]:
```
#Create subset2
df2=df[['Page total likes','Type','Category','Post Month','Post Weekday']].loc[16:
df2
```

Out[11]:

	Page total likes	Type	Category	Post Month	Post Weekday
16	138414	Photo	3	12	3
17	138414	Photo	1	12	2
18	138414	Status	3	12	2
19	138414	Photo	3	12	1
20	138414	Photo	2	12	1
21	138414	Photo	1	12	7
22	138414	Link	1	12	7
23	138414	Photo	3	12	7
24	138414	Status	2	12	6
25	138458	Status	2	12	6
26	138458	Status	2	12	5
27	138458	Photo	3	12	5
28	138895	Photo	2	12	5
29	138895	Video	1	12	4
30	138895	Photo	2	12	4

In [12]:

```
#Create subset3
df3=df[['Page total likes','Type','Category','Post Month','Post Weekday']].loc[31:
df3
```

Out[12]:

	Page total likes	Type	Category	Post Month	Post Weekday
31	138895	Photo	2	12	3
32	138895	Photo	3	12	3
33	138895	Photo	3	12	2
34	138895	Photo	1	12	2
35	138895	Photo	2	12	1
36	138895	Photo	3	12	1
37	138895	Photo	1	12	7
38	138895	Status	2	12	7
39	138895	Photo	1	12	7
40	138895	Status	2	12	6
41	138895	Link	1	12	6
42	138353	Photo	1	12	5
43	138353	Link	1	12	5
44	138353	Photo	1	12	4
45	138353	Link	1	12	4
46	138353	Status	1	12	3
47	138353	Link	1	12	3
48	138353	Photo	1	12	2
49	138353	Link	1	12	2
50	138353	Photo	2	11	1

In [13]:

```
#merging
merg=pd.concat([df1,df2,df3])
merg
```

Out[13]:

	Page total likes	Category	Post Month	Post Weekday	Type
0	139441	2	12	4	NaN
1	139441	2	12	3	NaN
2	139441	3	12	3	NaN
3	139441	2	12	2	NaN
4	139441	2	12	2	NaN
5	139441	2	12	1	NaN
6	139441	3	12	1	NaN
7	139441	3	12	7	NaN
8	139441	2	12	7	NaN
9	139441	3	12	6	NaN
10	139441	2	12	5	NaN
11	139441	2	12	5	NaN
12	139441	2	12	5	NaN
13	139441	2	12	5	NaN
14	138414	2	12	4	NaN
15	138414	2	12	3	NaN
16	138414	3	12	3	Photo
17	138414	1	12	2	Photo
18	138414	3	12	2	Status
19	138414	3	12	1	Photo
20	138414	2	12	1	Photo
21	138414	1	12	7	Photo
22	138414	1	12	7	Link
23	138414	3	12	7	Photo
24	138414	2	12	6	Status
25	138458	2	12	6	Status
26	138458	2	12	5	Status
27	138458	3	12	5	Photo
28	138895	2	12	5	Photo
29	138895	1	12	4	Video
30	138895	2	12	4	Photo
31	138895	2	12	3	Photo
32	138895	3	12	3	Photo
33	138895	3	12	2	Photo
34	138895	1	12	2	Photo
35	138895	2	12	1	Photo

	Page total likes	Category	Post Month	Post Weekday	Type
36	138895	3	12	1	Photo
37	138895	1	12	7	Photo
38	138895	2	12	7	Status
39	138895	1	12	7	Photo
40	138895	2	12	6	Status
41	138895	1	12	6	Link
42	138353	1	12	5	Photo
43	138353	1	12	5	Link
44	138353	1	12	4	Photo
45	138353	1	12	4	Link
46	138353	1	12	3	Status
47	138353	1	12	3	Link
48	138353	1	12	2	Photo
49	138353	1	12	2	Link
50	138353	2	11	1	Photo

In [16]:
```
#Sorting
sort_values=df.sort_values('Page total likes',ascending=False)
sort_values
```

Out[16]:

	Page total likes	Type	Category	Post Month	Post Weekday	Post Hour	Paid	Lifetime Post Total Reach	Lifetime Post Total Impressions	Lifetime Engaged Users	C
0	139441	Photo	2	12	4	3	0.0	2752	5091	178	
8	139441	Status	2	12	7	3	0.0	11844	22538	1530	
1	139441	Status	2	12	3	10	0.0	10460	19057	1457	
12	139441	Photo	2	12	5	10	0.0	2847	5133	193	
11	139441	Photo	2	12	5	10	0.0	3112	5590	208	
...	...	...	...	...	...	...	...	...	...	...	
495	85093	Photo	3	1	7	2	0.0	4684	7536	733	
496	81370	Photo	2	1	5	8	0.0	3480	6229	537	
497	81370	Photo	1	1	5	2	0.0	3778	7216	625	
498	81370	Photo	3	1	4	11	0.0	4156	7564	626	
499	81370	Photo	2	1	4	4	NaN	4188	7292	564	

500 rows × 19 columns

In [20]:

```
#transpose
transposing=df.transpose()
transposing
```

Out[20]:

	0	1	2	3	4	5	6	7	8	9
Page total likes	139441	139441	139441	139441	139441	139441	139441	139441	139441	139441
Type	Photo	Status	Photo	Photo	Photo	Status	Photo	Photo	Status	Photo
Category	2	2	3	2	2	2	3	3	2	3
Post Month	12	12	12	12	12	12	12	12	12	12
Post Weekday	4	3	3	2	2	1	1	7	7	6
Post Hour	3	10	3	10	3	9	3	9	3	10
Paid	0.0	0.0	0.0	1.0	0.0	0.0	1.0	1.0	0.0	0.0
Lifetime Post Total Reach	2752	10460	2413	50128	7244	10472	11692	13720	11844	4694
Lifetime Post Total Impressions	5091	19057	4373	87991	13594	20849	19479	24137	22538	8668
Lifetime Engaged Users	178	1457	177	2211	671	1191	481	537	1530	280
Lifetime Post Consumers	109	1361	113	790	410	1073	265	232	1407	183
Lifetime Post Consumptions	159	1674	154	1119	580	1389	364	305	1692	250
Lifetime Post Impressions by people who have liked your Page	3078	11710	2812	61027	6228	16034	15432	19728	15220	4309
Lifetime Post reach by people who like your Page	1640	6112	1503	32048	3200	7852	9328	11056	7912	2324
Lifetime People who have liked your Page and engaged with your post	119	1108	132	1386	396	1016	379	422	1250	199
comment	4	5	0	58	19	1	3	0	0	3
like	79.0	130.0	66.0	1572.0	325.0	152.0	249.0	325.0	161.0	113.0
share	17.0	29.0	14.0	147.0	49.0	33.0	27.0	14.0	31.0	26.0
Total Interactions	100	164	80	1777	393	186	279	339	192	142

19 rows × 500 columns

In [21]:

```
#shaping
df.shape
```

Out[21]: (500, 19)

In [22]:
```
#reshaping
pivot_table=pd.pivot_table(df,index=['Type','Category'],values='comment')
print(pivot_table)
```

```
                   comment
Type   Category
Link   1          2.900000
       2          2.000000
       3          2.000000
Photo  1          5.897297
       2         11.692308
       3          6.913333
Status 1          4.333333
       2          9.921053
       3          2.750000
Video  1         12.285714
```

In [23]:
```
#reshaping
pivot_table=pd.pivot_table(df,index=['Type','Category'],values='like')
print(pivot_table)
```

```
                      like
Type   Category
Link   1          75.650000
       2          32.000000
       3          68.000000
Photo  1         126.000000
       2         235.857143
       3         219.753333
Status 1         136.333333
       2         182.552632
       3         151.500000
Video  1         231.428571
```